

High-Risk Groups:

- **Haemodialysis** patients, **physicians, surgeons, dentists, paramedical staff, and laboratory/blood bank workers**
- Infected individuals are contagious during **incubation period** and as long as **HBsAg-positive**

Acute Hepatitis Clinical Features

1. Prodromal phase:

- **Anorexia, nausea, vomiting, headache, fatigue, malaise, myalgia, arthralgia**
- **Low-grade fever**

2. Icteric phase:

- **Jaundice** appears with dark urine and yellowish discoloration of skin and eyes
- **Clay-colored stool** and **pruritus** (suggests **cholestasis**)
- Some may never get jaundice (anicteric hepatitis)

3. Recovery phase:

- Improvement in symptoms and reduction in jaundice
- **Clinical and biochemical recovery** within 3-4 months

Signs of Acute Hepatitis:

- **Yellow sclera**
- **Skin scratch marks** due to **pruritus**
- **Tender hepatomegaly** in more than 50% cases
- **Splenomegaly** and **lymphadenopathy** in 10-20% cases

Complications of Acute Hepatitis:

- **Acute fulminant hepatitis**
- **Relapsing hepatitis**
- **Cholestatic hepatitis**
- **Chronic hepatitis** and **cirrhosis**
- **Aplastic anaemia**
- **Papular acrodermatitis**
- **Myelitis** and **neuropathy**

- **COPD** is a **heterogeneous condition** including **chronic bronchitis**, **chronic bronchiolitis**, and **emphysema**.

Clinical Features

- **Suspect COPD** in patients over **40 years** with symptoms like **cough**, **sputum production**, or **breathlessness**.
- **Cough**: Usually the **first symptom** with small amounts of **muroid sputum**.
- **Physical signs**: Nonspecific and poorly correlate with lung function.
- Presence of **pitting oedema** should be noted.
- **Phenotypes**:
 - **Pink Puffers**: Thin, breathless, normal PaCO_2 until late stages.
 - **Blue Bloaters**: Early **hypercapnia**, **oedema**, and **secondary polycythemia**.

Pathophysiology bronchial asthma

- **Causes:** Combination of **genetic** and **environmental factors**.
- **Early onset (extrinsic):**
 - Starts in **childhood**, with a **family history of atopy** (e.g., allergic rhinitis, eczema).
 - **Positive skin test, raised IgE levels.**
- **Late onset (intrinsic):**
 - Starts in **adulthood**, no family history of allergies.
 - **Negative skin test, normal IgE levels.**
- **Stimuli:** Allergic (e.g., dust, pollen) or non-allergic (e.g., cold air, exercise).
- **Bronchospasm:** Caused by allergen exposure leading to two-phase response:

- **Early reaction (Type I):** **Histamine, leukotrienes** cause immediate bronchoconstriction.
- **Late reaction (Type II):** **T-cell mediated inflammation**, involving **eosinophils**, leads to chronic bronchial inflammation.

Clinical Features

- **Episodic asthma:**
 - Sudden attacks of **dyspnoea, cough, and wheezing**; more common in children.
 - Episodes last **hours to days**; patients are asymptomatic between attacks.
- **Chronic asthma:**
 - Persistent symptoms in older adults, worse in the **early morning**.
 - May resemble **COPD**.
- **Examination findings:**
 - **Tachypnoea, tachycardia**, use of **accessory muscles**.
 - **Prolonged expiration**, audible **wheeze**.
- **Acute severe asthma** (life-threatening):
 - **Tachycardia, cyanosis, silent chest** on auscultation, **confusion or drowsiness**.

Clinical Features: Thrombocytopenia

1. **Skin manifestations:** Petechiae, purpura, ecchymoses
2. **Mucosal bleeding:** Nosebleeds (epistaxis), gum bleeding
3. **Excessive bleeding after trauma or surgery**
4. **Menorrhagia:** Heavy menstrual bleeding
5. **Hematuria:** Blood in urine

Investigations:

1. **Complete Blood Count (CBC):** Platelet count below normal
2. **Peripheral Blood Smear:** To assess platelet morphology
3. **Bone Marrow Examination:** For decreased production causes
4. **Coagulation Profile:** PT, aPTT, fibrinogen levels
5. **Specific Tests:** Antiplatelet antibodies, viral markers for HIV or hepatitis